Incidental Affect Image Test

2026-03-11

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Admin

```
#setwd(dirname(rstudioapi::getActiveDocumentContext()$path))
library(tidyverse)
library(lme4)
library(lmerTest)
library(emmeans)
library(brms)
library(performance)
library(bridgesampling)
library(patchwork)
source("https://raw.githubusercontent.com/vyqlua/avi_functions/refs/heads/main/ipsatize_fn.R")
conflicted::conflicts_prefer(dplyr::filter)
conflicted::conflicts_prefer(lmerTest::lmer)
```

Data cleaning

```
# Taiwan Data

d_TN <- read.csv("raw_data/TN_ImageTest.csv", encoding = "UTF-8") %>%
  dplyr::filter(id != "TEST",
                Status == 0,
                IPAddress != "128.252.48.41",
                Finished == 1,
                consent1 == 1) %>%
  distinct(IPAddress, .keep_all = TRUE) %>%
  distinct(email, .keep_all = TRUE) %>%
  dplyr::select(-IPAddress, -starts_with("Location"), -starts_with("Recipient"),
                -email, -id, -UserLanguage, -Q_DuplicateRespondent,
                -(agre_screener:consent1)) %>%
  dplyr::mutate(Culture = "TN") %>%
  dplyr::select(StartDate:DistributionChannel, Culture, everything())

### NOTE: in qualtrics, arousal and valence question labels are switched.

d_TN <- d_TN %>%
  dplyr::rename_with(
    ~ str_replace_all(.,
                      c("_arou" = "_VALE",
                        "_vale" = "_AROU"))) %>%
  dplyr::rename_with(
    ~ str_replace_all(.,
                      c("_AROU" = "_arou",
                        "_VALE" = "_vale")))
```

```
# European American Data

d_EA <- read.csv("raw_data/EA_ImageTest.csv", encoding = "UTF-8") %>%
  dplyr::filter(id != "TEST",
                Status == 0,
```

```r
                Finished == 1,
                consent1 == 1)  %>%
  distinct(IPAddress, .keep_all = TRUE) %>%
  distinct(id, .keep_all = TRUE) %>%
  dplyr::select(-IPAddress, -starts_with("Location"), -starts_with("Recipient"),
                -email, -id, -UserLanguage, -Q_DuplicateRespondent,
                -(prescreen_age:consent1)) %>%
  dplyr::mutate(Culture = "EA") %>%
  dplyr::select(StartDate:DistributionChannel, Culture, everything())
```

```r
# Merge

d <- rbind(d_EA, d_TN) %>%
  dplyr::mutate(
    dplyr::across(X1_neu_vale:age, as.numeric))%>%
  dplyr::rename_with(
    ~ str_replace_all(.,
                      c("1_10" = ".sleepy",
                        "1_1"  = ".enthusiastic",
                        "1_2"  = ".astonished",
                        "1_3"  = ".nervous",
                        "1_4"  = ".dull",
                        "1_5"  = ".quiet",
                        "1_6"  = ".relaxed",
                        "1_7"  = ".excited",
                        "1_8"  = ".surprised",
                        "1_9"  = ".elated",
                        "2_10" = ".fearful",
                        "2_1"  = ".still",
                        "2_2"  = ".lonely",
                        "2_3"  = ".strong",
                        "2_4"  = ".passive",
                        "2_5"  = ".content",
                        "2_6"  = ".sluggish",
                        "2_7"  = ".inactive",
                        "2_8"  = ".sad",
                        "2_9"  = ".euphoric",
                        "3_10" = ".serene",
                        "3_1"  = ".happy",
                        "3_2"  = ".idle",
                        "3_3"  = ".calm",
                        "3_4"  = ".unhappy",
                        "3_5"  = ".aroused",
                        "3_6"  = ".hostile",
                        "3_7"  = ".satisfied",
                        "3_8"  = ".rested",
                        "3_9"  = ".peaceful",
                        "4_1"  = ".noemotion",
                        "4_2"  = ".energetic",
                        "4_3"  = ".ashamed",
                        "4_4"  = ".disgusted",
                        "4_5"  = ".stressed",
                        "4_6"  = ".guilty",
```

```
                        "4_7"  = ".contemptuous",
                        "4_8"  = ".fatigued",
                        "4_9"  = ".angry")))

write.csv(d, "imageTestData.csv")

rm(d_EA, d_TN, d)
```

Load data

```
d <- read.csv("imageTestData.csv") %>% dplyr::select(-"X")
```

Sample Demographics

```
# European American
d %>% dplyr::filter(Culture == "EA") %>%
  summarize(Culture = "EA",
            M_age = mean(age, na.rm = T),
            SD_age = sd(age, na.rm = T),
            Female = sum(gender == "2", na.rm = T)/ n())
```

```
##   Culture    M_age   SD_age    Female
## 1      EA 44.76531 16.82401 0.7373737
```

```
# TN
d %>% dplyr::filter(Culture == "TN") %>%
  summarize(Culture = "TN",
            M_age = mean(age, na.rm = T),
            SD_age = sd(age, na.rm = T),
            Female = sum(gender == "2", na.rm = T)/ n())
```

```
##   Culture    M_age   SD_age    Female
## 1      TN 32.21296 9.833237 0.6513761
```

AVI

```
d <- d %>%
  ipsatize_avi("r.",
               remove = c("euphoric", "contemptuous", "angry"),
               full_avi = TRUE,
               maximizing_pos = TRUE)
```

```
## Items ipsatized successfully. Variables created:
```

```
##  [1] "rMean"             "rSd"               "r.enthusiastic_ip"
##  [4] "r.astonished_ip"   "r.nervous_ip"      "r.dull_ip"
##  [7] "r.quiet_ip"        "r.relaxed_ip"      "r.excited_ip"
## [10] "r.surprised_ip"    "r.elated_ip"       "r.sleepy_ip"
## [13] "r.still_ip"        "r.lonely_ip"       "r.strong_ip"
## [16] "r.passive_ip"      "r.content_ip"      "r.sluggish_ip"
## [19] "r.inactive_ip"     "r.sad_ip"          "r.euphoric_ip"
## [22] "r.fearful_ip"      "r.happy_ip"        "r.idle_ip"
## [25] "r.calm_ip"         "r.unhappy_ip"      "r.aroused_ip"
## [28] "r.hostile_ip"      "r.satisfied_ip"    "r.rested_ip"
## [31] "r.peaceful_ip"     "r.serene_ip"       "r.noemotion_ip"
## [34] "r.energetic_ip"    "r.ashamed_ip"      "r.disgusted_ip"
## [37] "r.stressed_ip"     "r.guilty_ip"       "r.contemptuous_ip"
## [40] "r.fatigued_ip"     "r.angry_ip"
```

```
## AVI Variables Computed:
```

```
##  [1] "rHAP"       "rLAP"       "rHAN"       "rLAN"       "rHAP_ip"
##  [6] "rLAP_ip"    "rHAN_ip"    "rLAN_ip"    "rPOS"       "rNEG"
## [11] "rPOS_ip"    "rNEG_ip"    "rMaxPos"    "rMaxPos_ip"
```

```
## Check 'avi_codebook' in environment to check how each AVI variable was calculated.
```

```r
d <- d %>%
  ipsatize_avi("i.",
               remove = c("euphoric", "contemptuous", "angry"),
               full_avi = TRUE,
               maximizing_pos = TRUE)
```

```
## Items ipsatized successfully. Variables created:
```

```
##  [1] "iMean"             "iSd"               "i.enthusiastic_ip"
##  [4] "i.astonished_ip"   "i.nervous_ip"      "i.dull_ip"
##  [7] "i.quiet_ip"        "i.relaxed_ip"      "i.excited_ip"
## [10] "i.surprised_ip"    "i.elated_ip"       "i.sleepy_ip"
## [13] "i.still_ip"        "i.lonely_ip"       "i.strong_ip"
## [16] "i.passive_ip"      "i.content_ip"      "i.sluggish_ip"
## [19] "i.inactive_ip"     "i.sad_ip"          "i.fearful_ip"
## [22] "i.happy_ip"        "i.idle_ip"         "i.calm_ip"
## [25] "i.unhappy_ip"      "i.aroused_ip"      "i.hostile_ip"
## [28] "i.satisfied_ip"    "i.rested_ip"       "i.peaceful_ip"
## [31] "i.serene_ip"       "i.noemotion_ip"    "i.energetic_ip"
## [34] "i.ashamed_ip"      "i.disgusted_ip"    "i.stressed_ip"
## [37] "i.guilty_ip"       "i.fatigued_ip"     "i.euphoric_ip"
## [40] "i.contemptuous_ip" "i.angry_ip"
```

```
## AVI Variables Computed:
```

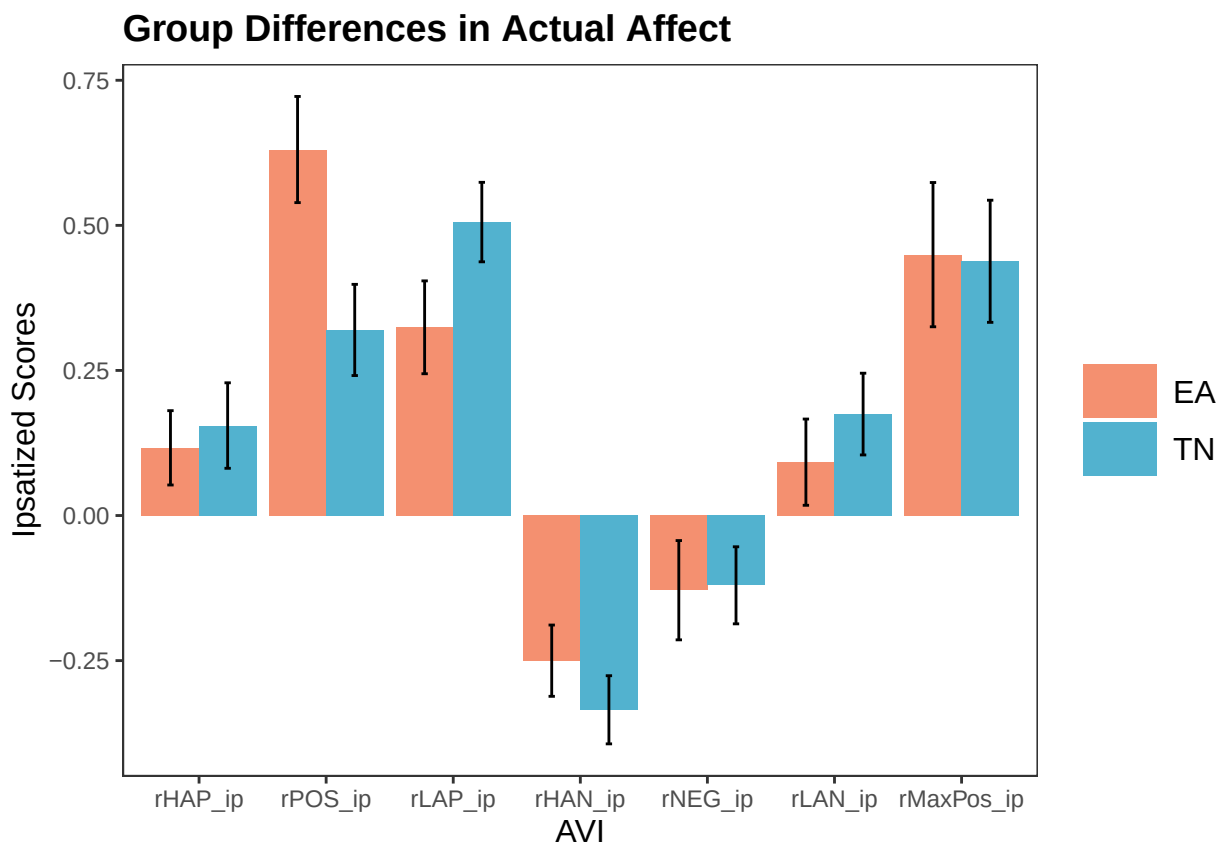
```
##  [1] "iHAP"       "iLAP"       "iHAN"       "iLAN"       "iHAP_ip"
##  [6] "iLAP_ip"    "iHAN_ip"    "iLAN_ip"    "iPOS"       "iNEG"
## [11] "iPOS_ip"    "iNEG_ip"    "iMaxPos"    "iMaxPos_ip"
```

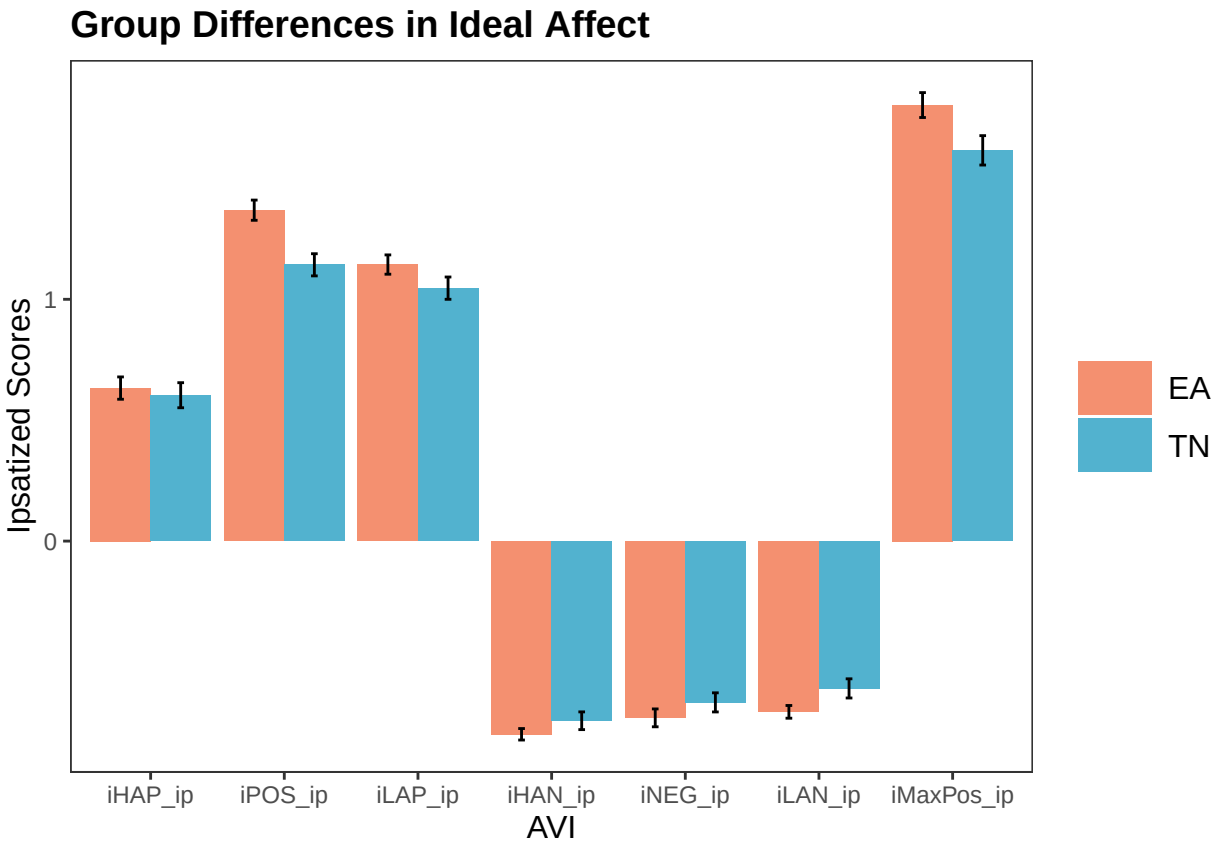
```
## Check 'avi_codebook' in environment to check how each AVI variable was calculated.
```

```r
plot_avi(
  d %>% dplyr::select(Culture, rHAP:rMaxPos_ip, iHAP:iMaxPos_ip),
  "Culture")
```

```
## Warning: Using an external vector in selections was deprecated in tidyselect 1.1.0.
## i Please use `all_of()` or `any_of()` instead.
##   # Was:
##   data %>% select(group_id)
##
##   # Now:
##   data %>% select(all_of(group_id))
##
## See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## `summarise()` has grouped output by 'Culture'. You can override using the
## `.groups` argument.
```




Group Differences in Ideal Affect
Ipsatized Scores
1
0
iHAP_ip
iPOS_ip
iLAP_ip
iHAN_ip
iNEG_ip
iLAN_ip
iMaxPos_ip
AVI
EA
TN

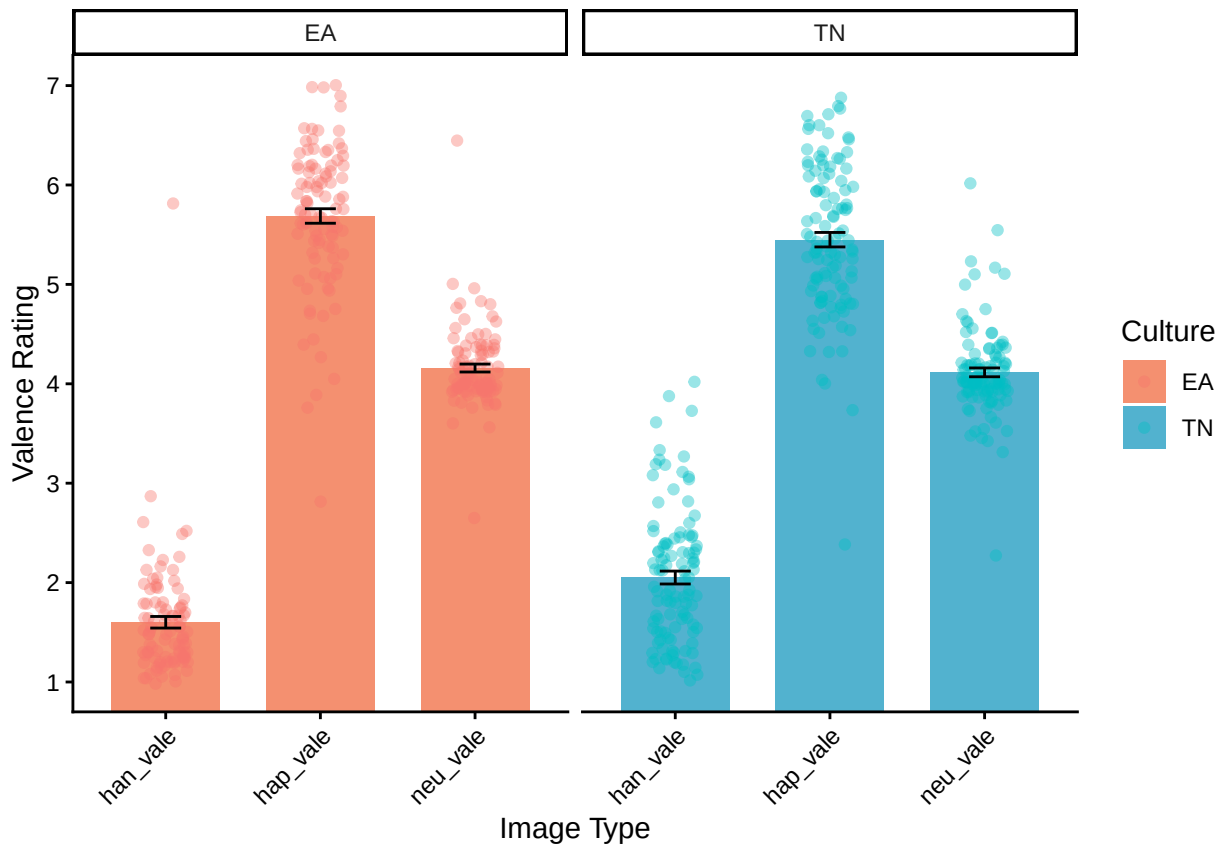
Continuous Valence & Arousal

```
d <- d %>% dplyr::mutate(
  han_vale = dplyr::across(contains("_han_vale")) %>% rowMeans(),
  han_arou = dplyr::across(contains("_han_arou")) %>% rowMeans(),
  hap_vale = dplyr::across(contains("_hap_vale")) %>% rowMeans(),
  hap_arou = dplyr::across(contains("_hap_arou")) %>% rowMeans(),
  neu_vale = dplyr::across(contains("_neu_vale")) %>% rowMeans(),
  neu_arou = dplyr::across(contains("_neu_arou")) %>% rowMeans())
```

Valence

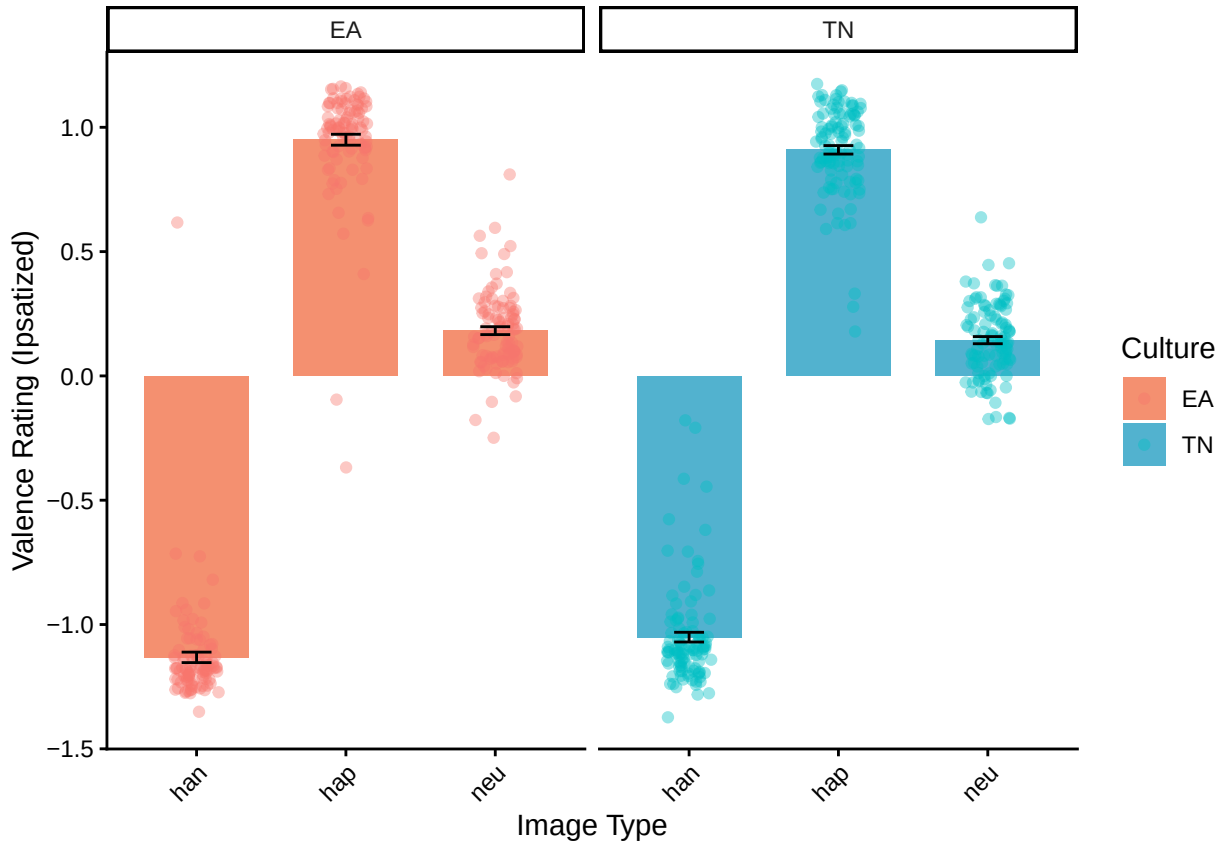
Plot

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



```
## `summarise()` has grouped output by 'ResponseId', 'Culture'. You can override
## using the `.groups` argument.
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



By Image Type

```r
# create long valence data for analysis

d_vale <- d %>%
  dplyr::select(ResponseId, Culture,
                contains("_vale"),
                -hap_vale, -han_vale, -neu_vale)

d_vale_long <- d_vale %>%
  pivot_longer(
    cols = -c(ResponseId, Culture),
    names_to = "dimension",
    values_to = "value"
  ) %>%
  dplyr::mutate(
    image_no = str_extract(dimension, "(?<=^X)[0-9]+"),
    affect = substr(dimension, nchar(dimension)-7, nchar(dimension)-5))

# ipsatize valence ratings

d_vale_long <- d_vale_long %>%
  group_by(ResponseId) %>%
  dplyr::mutate(
    value_ip = (value - mean(value, na.rm = T)) / sd(value, na.rm = T)) %>%
  ungroup()
```

```r
# non-ip
rating_by_imagetype <-
  lmer(value ~ affect + (1 | ResponseId) + (1 | image_no),
       data = d_vale_long,
       REML = FALSE)

summary(rating_by_imagetype)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: value ~ affect + (1 | ResponseId) + (1 | image_no)
##    Data: d_vale_long
##
##      AIC      BIC   logLik deviance df.resid
##  28520.9  28564.1 -14254.4  28508.9     9974
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -5.3405 -0.5863 -0.0510  0.5046  5.5067
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  ResponseId (Intercept) 0.08846  0.2974
##  image_no   (Intercept) 0.09517  0.3085
##  Residual               0.96955  0.9847
```

```
## Number of obs: 9980, groups:  ResponseId, 208; image_no, 48
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)  1.83810    0.08164 54.64887   22.52   <2e-16 ***
## affecthap    3.72702    0.11171 47.98156   33.36   <2e-16 ***
## affectneu    2.29714    0.11171 47.98225   20.56   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) affcth
## affecthap -0.684
## affectneu -0.684  0.500
```

```r
emmeans(rating_by_imagetype, pairwise ~ affect)
```

```
## $emmeans
##  affect emmean     SE  df asymp.LCL asymp.UCL
##  han      1.84 0.0816 Inf      1.68      2.00
##  hap      5.57 0.0816 Inf      5.41      5.73
##  neu      4.14 0.0816 Inf      3.98      4.30
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## $contrasts
##  contrast  estimate    SE  df z.ratio p.value
##  han - hap    -3.73 0.112 Inf -33.364  <.0001
##  han - neu    -2.30 0.112 Inf -20.563  <.0001
##  hap - neu     1.43 0.112 Inf  12.800  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
#ip
rating_by_imagetype_ip <-
  lmer(value_ip ~ affect + (1 | image_no),
       data = d_vale_long,
       REML = FALSE) # removed (1 | responseID) since its ipsatized

summary(rating_by_imagetype_ip)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: value_ip ~ affect + (1 | image_no)
##    Data: d_vale_long
##
##      AIC      BIC   logLik deviance df.resid
##  14793.8  14829.9  -7391.9  14783.8     9927
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -6.0911 -0.4788 -0.0599  0.4841  7.6829
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  image_no (Intercept) 0.03013  0.1736
##  Residual             0.25538  0.5054
## Number of obs: 9932, groups:  image_no, 48
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept) -1.09096    0.04427 48.00052  -24.64   <2e-16 ***
## affecthap    2.02075    0.06261 48.00052   32.27   <2e-16 ***
## affectneu    1.25206    0.06261 48.00109   20.00   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) affcth
## affecthap -0.707
## affectneu -0.707  0.500
```

```r
emmeans(rating_by_imagetype_ip, pairwise ~ affect)
```

```
## $emmeans
##  affect emmean     SE  df asymp.LCL asymp.UCL
##  han    -1.091 0.0443 Inf   -1.1777    -1.004
##  hap     0.930 0.0443 Inf    0.8430     1.017
##  neu     0.161 0.0443 Inf    0.0743     0.248
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## $contrasts
##  contrast  estimate     SE  df z.ratio p.value
##  han - hap   -2.021 0.0626 Inf -32.273  <.0001
##  han - neu   -1.252 0.0626 Inf -19.997  <.0001
##  hap - neu    0.769 0.0626 Inf  12.277  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: tukey method for comparing a family of 3 estimates
```

By Image Type * Culture

```r
#non-ip

rating_by_imagetype_culture <-
  lmer(value ~ affect * Culture + (1 | ResponseId) + (1 | image_no),
       data = d_vale_long,
       REML = FALSE)

summary(rating_by_imagetype_culture)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: value ~ affect * Culture + (1 | ResponseId) + (1 | image_no)
##    Data: d_vale_long
##
##      AIC      BIC   logLik deviance df.resid
##  28307.2  28372.1 -14144.6  28289.2     9971
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -5.5545 -0.5458 -0.0594  0.5273  5.7871
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  ResponseId (Intercept) 0.08807  0.2968
##  image_no   (Intercept) 0.09529  0.3087
##  Residual               0.94805  0.9737
## Number of obs: 9980, groups:  ResponseId, 208; image_no, 48
##
## Fixed effects:
##                       Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)            1.60101    0.08628   67.75879  18.557  < 2e-16 ***
## affecthap              4.08712    0.11449   52.91587  35.698  < 2e-16 ***
## affectneu              2.55745    0.11449   52.91587  22.338  < 2e-16 ***
## CultureTN              0.45255    0.05329  386.08247   8.492 4.44e-16 ***
## affecthap:CultureTN   -0.68736    0.04780 9724.94339 -14.380  < 2e-16 ***
## affectneu:CultureTN   -0.49691    0.04780 9724.95438 -10.395  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) affcth affctn CltrTN affcth:CTN
## affecthap   -0.664
## affectneu   -0.664  0.500
## CultureTN   -0.324  0.098  0.098
## affcthp:CTN  0.145 -0.219 -0.109 -0.448
## affctn:ClTN  0.145 -0.109 -0.219 -0.448  0.500
```

```r
emmeans(rating_by_imagetype_culture, pairwise ~ affect | Culture)
```

```
## $emmeans
```

```
## Culture = EA:
##  affect emmean     SE  df asymp.LCL asymp.UCL
##  han      1.60 0.0863 Inf      1.43      1.77
##  hap      5.69 0.0863 Inf      5.52      5.86
##  neu      4.16 0.0863 Inf      3.99      4.33
##
## Culture = TN:
##  affect emmean     SE  df asymp.LCL asymp.UCL
##  han      2.05 0.0855 Inf      1.89      2.22
##  hap      5.45 0.0855 Inf      5.29      5.62
##  neu      4.11 0.0855 Inf      3.95      4.28
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## $contrasts
## Culture = EA:
##  contrast  estimate    SE  df z.ratio p.value
##  han - hap    -4.09 0.114 Inf -35.698  <.0001
##  han - neu    -2.56 0.114 Inf -22.338  <.0001
##  hap - neu     1.53 0.114 Inf  13.361  <.0001
##
## Culture = TN:
##  contrast  estimate    SE  df z.ratio p.value
##  han - hap    -3.40 0.114 Inf -29.819  <.0001
##  han - neu    -2.06 0.114 Inf -18.073  <.0001
##  hap - neu     1.34 0.114 Inf  11.746  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
emmeans(rating_by_imagetype_culture, pairwise ~ Culture | affect)
```

```
## $emmeans
## affect = han:
##  Culture emmean     SE  df asymp.LCL asymp.UCL
##  EA        1.60 0.0863 Inf      1.43      1.77
##  TN        2.05 0.0855 Inf      1.89      2.22
##
## affect = hap:
##  Culture emmean     SE  df asymp.LCL asymp.UCL
##  EA        5.69 0.0863 Inf      5.52      5.86
##  TN        5.45 0.0855 Inf      5.29      5.62
##
## affect = neu:
##  Culture emmean     SE  df asymp.LCL asymp.UCL
##  EA        4.16 0.0863 Inf      3.99      4.33
##  TN        4.11 0.0855 Inf      3.95      4.28
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## $contrasts
## affect = han:
```

```
##  contrast estimate     SE  df z.ratio p.value
##  EA - TN   -0.4525 0.0533 Inf  -8.492  <.0001
##
## affect = hap:
##  contrast estimate     SE  df z.ratio p.value
##  EA - TN    0.2348 0.0533 Inf   4.406  <.0001
##
## affect = neu:
##  contrast estimate     SE  df z.ratio p.value
##  EA - TN    0.0444 0.0533 Inf   0.832  0.4052
##
## Degrees-of-freedom method: asymptotic
```

```r
print("nonip ratings"); r2(rating_by_imagetype); r2(rating_by_imagetype_culture)
```

```
## [1] "nonip ratings"

## # R2 for Mixed Models
##
##   Conditional R2: 0.724
##      Marginal R2: 0.672

## # R2 for Mixed Models
##
##   Conditional R2: 0.730
##      Marginal R2: 0.678
```

```r
# ip
rating_by_imagetype_culture_ip <-
  lmer(value_ip ~ affect * Culture + (1 | image_no),
       data = d_vale_long,
       REML = FALSE) # removed (1 | responseID) since its ipsatized

summary(rating_by_imagetype_culture_ip)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: value_ip ~ affect * Culture + (1 | image_no)
##    Data: d_vale_long
##
##      AIC      BIC   logLik deviance df.resid
##  14769.5  14827.2  -7376.8  14753.5     9924
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.1753 -0.4865 -0.0519  0.4855  7.6198
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  image_no (Intercept) 0.03013  0.1736
##  Residual             0.25460  0.5046
## Number of obs: 9932, groups:  image_no, 48
##
## Fixed effects:
##                       Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)           -1.13212    0.04521   52.19346 -25.040  < 2e-16 ***
## affecthap              2.08219    0.06394   52.19346  32.565  < 2e-16 ***
## affectneu              1.31419    0.06394   52.19346  20.553  < 2e-16 ***
## CultureTN              0.07892    0.01755 9884.00200   4.496 7.00e-06 ***
## affecthap:CultureTN   -0.11779    0.02483 9884.00200  -4.745 2.12e-06 ***
## affectneu:CultureTN   -0.11912    0.02483 9884.00242  -4.798 1.63e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) affcth affctn CltrTN affcth:CTN
## affecthap   -0.707
## affectneu   -0.707  0.500
## CultureTN   -0.203  0.143  0.143
## affcthp:CTN  0.143 -0.203 -0.101 -0.707
## affctn:ClTN  0.143 -0.101 -0.203 -0.707  0.500
```

```r
# emmeans(rating_by_imagetype_culture_ip, pairwise ~ affect | Culture)
emmeans(rating_by_imagetype_culture_ip, pairwise ~ Culture | affect)
```

```
## $emmeans
## affect = han:
##  Culture emmean     SE  df asymp.LCL asymp.UCL
##  EA      -1.132 0.0452 Inf   -1.2207    -1.044
```

```
##  TN      -1.053 0.0451 Inf   -1.1415    -0.965
##
## affect = hap:
##  Culture emmean     SE  df asymp.LCL asymp.UCL
##  EA       0.950 0.0452 Inf    0.8614     1.039
##  TN       0.911 0.0451 Inf    0.8229     1.000
##
## affect = neu:
##  Culture emmean     SE  df asymp.LCL asymp.UCL
##  EA       0.182 0.0452 Inf    0.0934     0.271
##  TN       0.142 0.0451 Inf    0.0535     0.230
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## $contrasts
## affect = han:
##  contrast estimate     SE  df z.ratio p.value
##  EA - TN   -0.0789 0.0176 Inf  -4.496  <.0001
##
## affect = hap:
##  contrast estimate     SE  df z.ratio p.value
##  EA - TN    0.0389 0.0176 Inf   2.214  0.0268
##
## affect = neu:
##  contrast estimate     SE  df z.ratio p.value
##  EA - TN    0.0402 0.0176 Inf   2.289  0.0221
##
## Degrees-of-freedom method: asymptotic
```

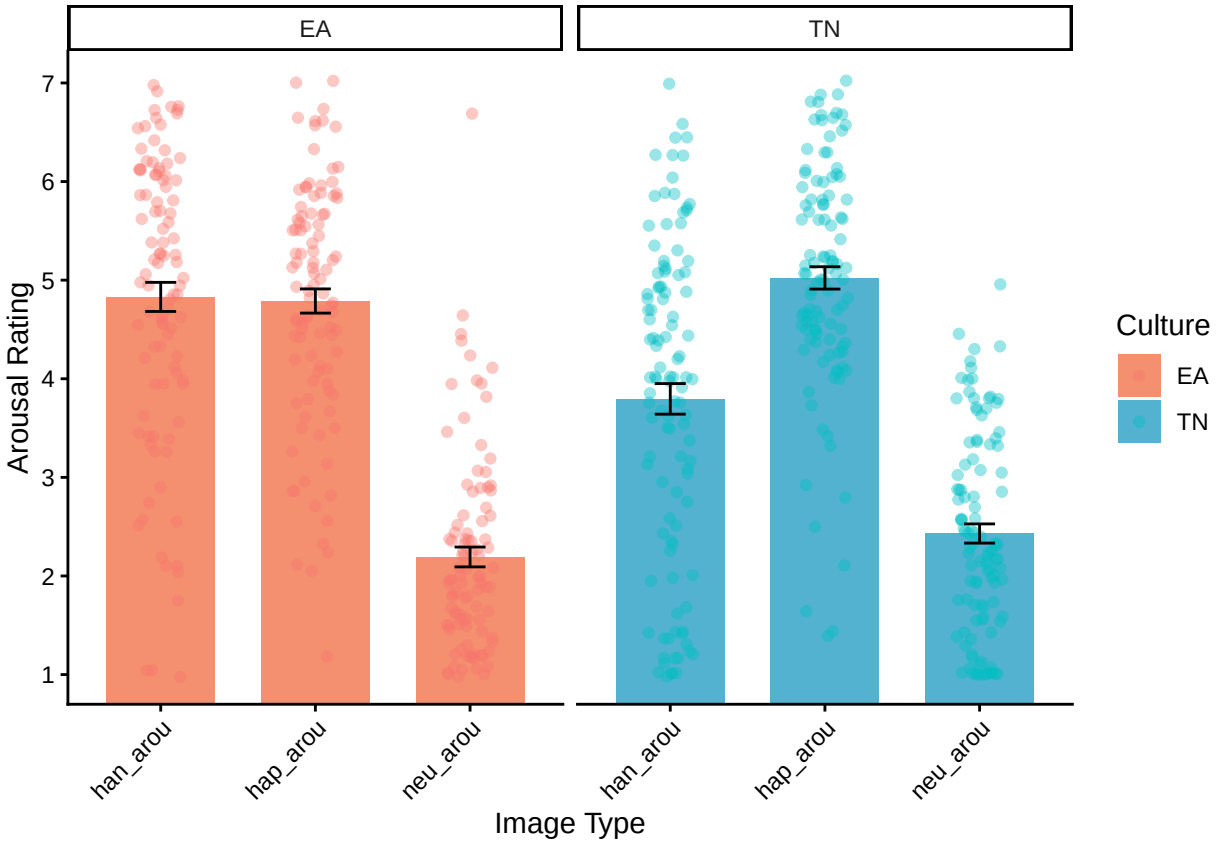
```r
print("ip ratings"); r2(rating_by_imagetype_ip); r2(rating_by_imagetype_culture_ip)
```

```
## [1] "ip ratings"

## # R2 for Mixed Models
##
##   Conditional R2: 0.739
##      Marginal R2: 0.708

## # R2 for Mixed Models
##
##   Conditional R2: 0.740
##      Marginal R2: 0.709
```

Arousal

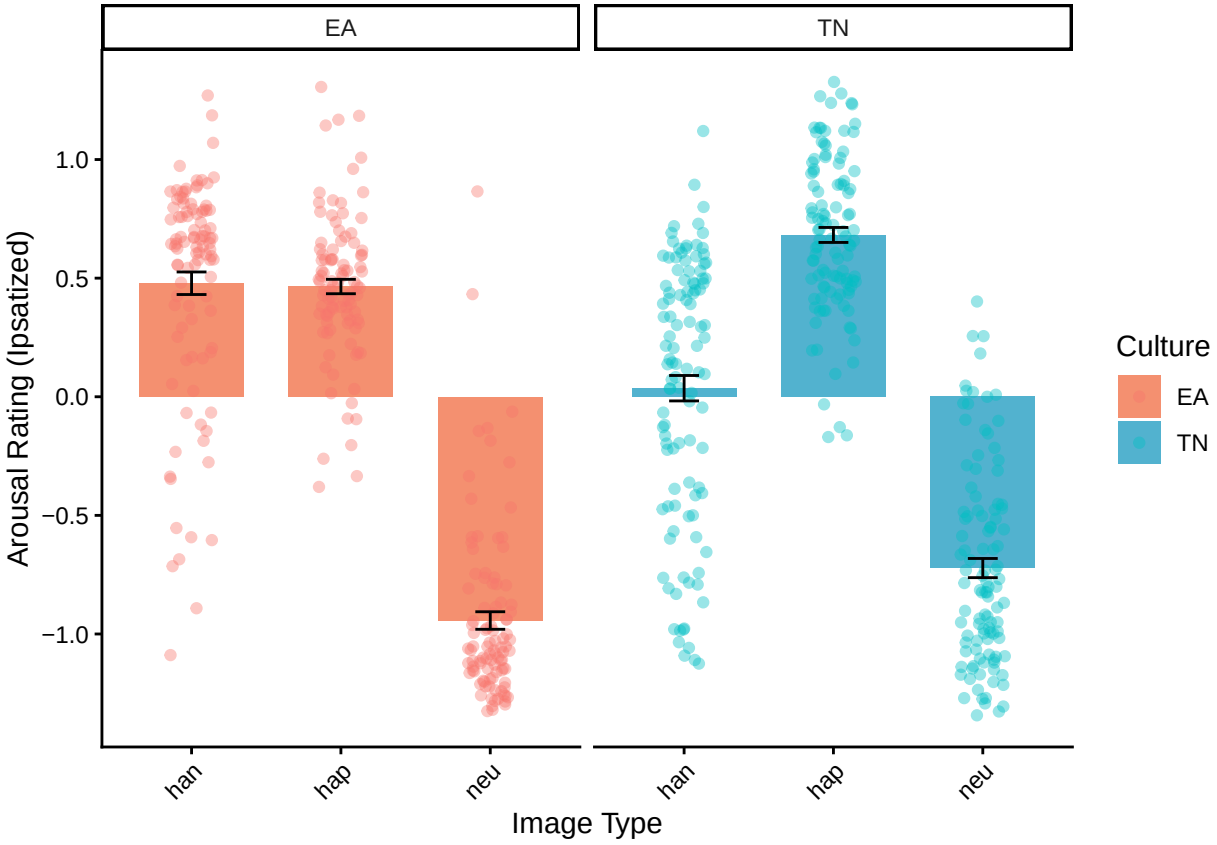
Plot

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



```
## `summarise()` has grouped output by 'ResponseId', 'Culture'. You can override
## using the `.groups` argument.
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



By Image Type

```
d_arou <- d %>%
  dplyr::select(ResponseId, Culture,
                contains("_arou"),
                -hap_arou, -han_arou, -neu_arou)

d_arou_long <- d_arou %>%
  pivot_longer(
    cols = -c(ResponseId, Culture),
    names_to = "dimension",
    values_to = "value"
  ) %>%
  dplyr::mutate(
    image_no = str_extract(dimension, "(?<=^X)[0-9]+"),
    affect = substr(dimension, nchar(dimension)-7, nchar(dimension)-5))

# ipsatize arousal ratings

d_arou_long <- d_arou_long %>%
  group_by(ResponseId) %>%
  dplyr::mutate(
    value_ip = (value - mean(value, na.rm = T)) / sd(value, na.rm = T)) %>%
  ungroup()
```

```
arousal_by_imagetype <-
  lmer(value ~ affect + (1 | ResponseId) + (1 | image_no),
       data = d_arou_long)

summary(arousal_by_imagetype)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ affect + (1 | ResponseId) + (1 | image_no)
##    Data: d_arou_long
##
## REML criterion at convergence: 36109.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.5216 -0.6785  0.0325  0.7258  4.0648
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  ResponseId (Intercept) 0.80067  0.8948
##  image_no   (Intercept) 0.05632  0.2373
##  Residual               2.02981  1.4247
## Number of obs: 9981, groups:  ResponseId, 208; image_no, 48
##
## Fixed effects:
##              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)   4.28816    0.08933 138.88830  48.005  < 2e-16 ***
```

```
## affecthap     0.62686    0.09089  44.99931   6.897 1.45e-08 ***
## affectneu    -1.96647    0.09089  45.00135 -21.636  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) affcth
## affecthap -0.509
## affectneu -0.509  0.500
```

```r
emmeans(arousal_by_imagetype, pairwise ~ affect)
```

```
## $emmeans
##  affect emmean     SE  df asymp.LCL asymp.UCL
##  han      4.29 0.0893 Inf      4.11      4.46
##  hap      4.92 0.0893 Inf      4.74      5.09
##  neu      2.32 0.0893 Inf      2.15      2.50
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## $contrasts
##  contrast  estimate     SE  df z.ratio p.value
##  han - hap   -0.627 0.0909 Inf  -6.897  <.0001
##  han - neu    1.966 0.0909 Inf  21.636  <.0001
##  hap - neu    2.593 0.0909 Inf  28.533  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
arousal_by_imagetype_ip <-
  lmer(value_ip ~ affect + (1 | image_no),
       data = d_arou_long)

summary(arousal_by_imagetype_ip)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value_ip ~ affect + (1 | image_no)
##    Data: d_arou_long
##
## REML criterion at convergence: 23355.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -7.5890 -0.6705  0.0262  0.7043  7.9837
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  image_no (Intercept) 0.01999  0.1414
##  Residual             0.60110  0.7753
## Number of obs: 9981, groups:  image_no, 48
##
```

```
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)  0.24665    0.03781 44.99829   6.523 5.21e-08 ***
## affecthap    0.33253    0.05347 45.00001   6.218 1.48e-07 ***
## affectneu   -1.07275    0.05347 45.00172 -20.061  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) affcth
## affecthap -0.707
## affectneu -0.707  0.500
```

```r
emmeans(arousal_by_imagetype_ip, pairwise ~ affect)
```

```
## $emmeans
##  affect emmean     SE  df asymp.LCL asymp.UCL
##  han     0.247 0.0378 Inf     0.173     0.321
##  hap     0.579 0.0378 Inf     0.505     0.653
##  neu    -0.826 0.0378 Inf    -0.900    -0.752
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## $contrasts
##  contrast  estimate     SE  df z.ratio p.value
##  han - hap   -0.333 0.0535 Inf  -6.218  <.0001
##  han - neu    1.073 0.0535 Inf  20.061  <.0001
##  hap - neu    1.405 0.0535 Inf  26.279  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: tukey method for comparing a family of 3 estimates
```

By Image Type * Culture

```r
arousal_by_imagetype_culture <-
  lmer(value ~ affect * Culture + (1 | ResponseId) + (1 | image_no),
       data = d_arou_long)

summary(arousal_by_imagetype_culture)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ affect * Culture + (1 | ResponseId) + (1 | image_no)
##    Data: d_arou_long
##
## REML criterion at convergence: 35661.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.9325 -0.6995  0.0442  0.7008  4.1564
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  ResponseId (Intercept) 0.79843  0.8935
##  image_no   (Intercept) 0.05679  0.2383
##  Residual               1.93722  1.3918
## Number of obs: 9981, groups:  ResponseId, 208; image_no, 48
##
## Fixed effects:
##                       Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)            4.83018    0.11330  226.42063  42.631  < 2e-16 ***
## affecthap             -0.04167    0.09770   60.03473  -0.426    0.671
## affectneu             -2.63763    0.09770   60.03473 -26.998  < 2e-16 ***
## CultureTN             -1.03431    0.13313  247.50195  -7.769  2.1e-13 ***
## affecthap:CultureTN    1.27584    0.06832 9724.00015  18.673  < 2e-16 ***
## affectneu:CultureTN    1.28098    0.06833 9724.00440  18.747  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) affcth affctn CltrTN affcth:CTN
## affecthap   -0.431
## affectneu   -0.431  0.500
## CultureTN   -0.616  0.094  0.094
## affcthp:CTN  0.158 -0.366 -0.183 -0.257
## affctn:ClTN  0.158 -0.183 -0.366 -0.257  0.500
```

```r
emmeans(arousal_by_imagetype_culture, pairwise ~ affect | Culture)
```

```
## Note: D.f. calculations have been disabled because the number of observations exceeds 3000.
## To enable adjustments, add the argument 'pbkrtest.limit = 9981' (or larger)
## [or, globally, 'set emm_options(pbkrtest.limit = 9981)' or larger];
## but be warned that this may result in large computation time and memory use.
```

```
## Note: D.f. calculations have been disabled because the number of observations exceeds 3000.
## To enable adjustments, add the argument 'lmerTest.limit = 9981' (or larger)
## [or, globally, 'set emm_options(lmerTest.limit = 9981)' or larger];
## but be warned that this may result in large computation time and memory use.

## $emmeans
## Culture = EA:
##  affect emmean    SE  df asymp.LCL asymp.UCL
##  han      4.83 0.113 Inf      4.61      5.05
##  hap      4.79 0.113 Inf      4.57      5.01
##  neu      2.19 0.113 Inf      1.97      2.41
##
## Culture = TN:
##  affect emmean    SE  df asymp.LCL asymp.UCL
##  han      3.80 0.109 Inf      3.58      4.01
##  hap      5.03 0.109 Inf      4.82      5.24
##  neu      2.44 0.109 Inf      2.22      2.65
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## $contrasts
## Culture = EA:
##  contrast  estimate     SE  df z.ratio p.value
##  han - hap   0.0417 0.0977 Inf   0.426  0.9046
##  han - neu   2.6376 0.0977 Inf  26.998  <.0001
##  hap - neu   2.5960 0.0977 Inf  26.572  <.0001
##
## Culture = TN:
##  contrast  estimate     SE  df z.ratio p.value
##  han - hap  -1.2342 0.0965 Inf -12.784  <.0001
##  han - neu   1.3566 0.0965 Inf  14.052  <.0001
##  hap - neu   2.5908 0.0966 Inf  26.834  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
emmeans(arousal_by_imagetype_culture, pairwise ~ Culture | affect)
```

```
## Note: D.f. calculations have been disabled because the number of observations exceeds 3000.
## To enable adjustments, add the argument 'pbkrtest.limit = 9981' (or larger)
## [or, globally, 'set emm_options(pbkrtest.limit = 9981)' or larger];
## but be warned that this may result in large computation time and memory use.
## Note: D.f. calculations have been disabled because the number of observations exceeds 3000.
## To enable adjustments, add the argument 'lmerTest.limit = 9981' (or larger)
## [or, globally, 'set emm_options(lmerTest.limit = 9981)' or larger];
## but be warned that this may result in large computation time and memory use.

## $emmeans
## affect = han:
##  Culture emmean    SE  df asymp.LCL asymp.UCL
##  EA        4.83 0.113 Inf      4.61      5.05
##  TN        3.80 0.109 Inf      3.58      4.01
```

```
##
## affect = hap:
##  Culture emmean    SE  df asymp.LCL asymp.UCL
##  EA        4.79 0.113 Inf      4.57      5.01
##  TN        5.03 0.109 Inf      4.82      5.24
##
## affect = neu:
##  Culture emmean    SE  df asymp.LCL asymp.UCL
##  EA        2.19 0.113 Inf      1.97      2.41
##  TN        2.44 0.109 Inf      2.22      2.65
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## $contrasts
## affect = han:
##  contrast estimate    SE  df z.ratio p.value
##  EA - TN     1.034 0.133 Inf   7.769  <.0001
##
## affect = hap:
##  contrast estimate    SE  df z.ratio p.value
##  EA - TN    -0.242 0.133 Inf  -1.814  0.0696
##
## affect = neu:
##  contrast estimate    SE  df z.ratio p.value
##  EA - TN    -0.247 0.133 Inf  -1.853  0.0639
##
## Degrees-of-freedom method: asymptotic
```

```r
print("nonip arousals"); r2(arousal_by_imagetype); r2(arousal_by_imagetype_culture)
```

```
## [1] "nonip arousals"
```

```
## # R2 for Mixed Models
##
##   Conditional R2: 0.506
##      Marginal R2: 0.297
```

```
## # R2 for Mixed Models
##
##   Conditional R2: 0.529
##      Marginal R2: 0.321
```

```r
arousal_by_imagetype_culture_ip <-
  lmer(value_ip ~ affect * Culture + (1 | image_no),
       data = d_arou_long)

summary(arousal_by_imagetype_culture_ip)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value_ip ~ affect * Culture + (1 | image_no)
##    Data: d_arou_long
##
## REML criterion at convergence: 22957.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -7.8872 -0.6626  0.0283  0.6702  8.3011
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  image_no (Intercept) 0.02011  0.1418
##  Residual             0.57675  0.7594
## Number of obs: 9981, groups:  image_no, 48
##
## Fixed effects:
##                       Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)            0.47853    0.04026   57.80306  11.886   <2e-16 ***
## affecthap             -0.01380    0.05694   57.80306  -0.242    0.809
## affectneu             -1.42179    0.05694   57.80306 -24.972   <2e-16 ***
## CultureTN             -0.44249    0.02636 9930.00165 -16.787   <2e-16 ***
## affecthap:CultureTN    0.66094    0.03728 9930.00314  17.729   <2e-16 ***
## affectneu:CultureTN    0.66619    0.03728 9930.00444  17.869   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) affcth affctn CltrTN affcth:CTN
## affecthap   -0.707
## affectneu   -0.707  0.500
## CultureTN   -0.343  0.243  0.243
## affcthp:CTN  0.243 -0.343 -0.172 -0.707
## affctn:ClTN  0.243 -0.172 -0.343 -0.707  0.500
```

```r
#emmeans(arousal_by_imagetype_culture_ip, pairwise ~ affect | Culture)
emmeans(arousal_by_imagetype_culture_ip, pairwise ~ Culture | affect)
```

```
## $emmeans
## affect = han:
##  Culture emmean     SE  df asymp.LCL asymp.UCL
##  EA       0.479 0.0403 Inf     0.400     0.557
##  TN       0.036 0.0398 Inf    -0.042     0.114
##
## affect = hap:
```

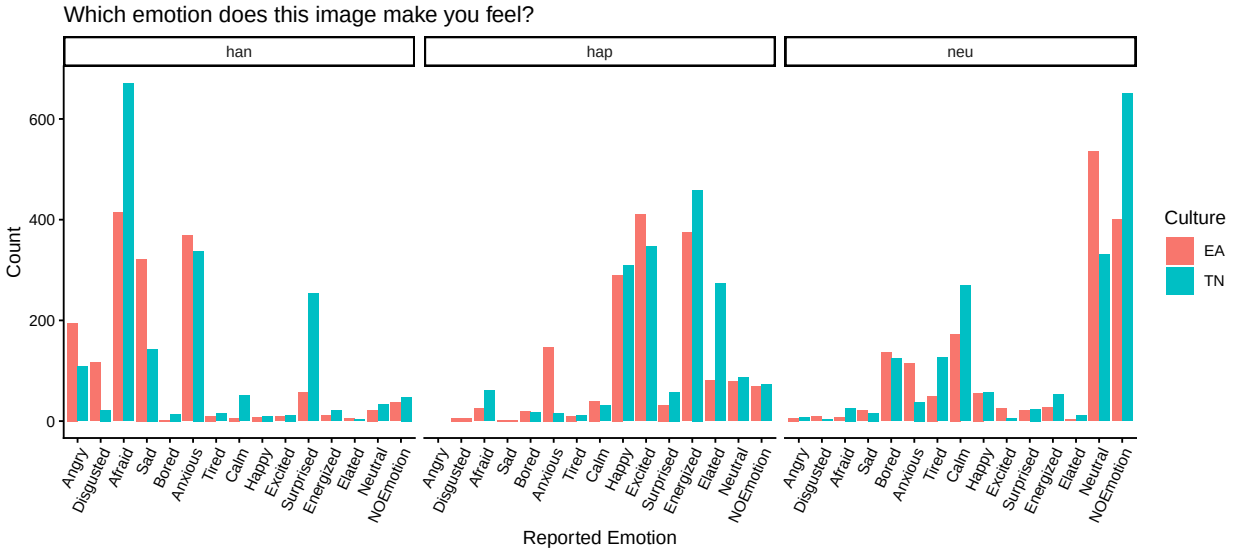
```
##  Culture emmean     SE  df asymp.LCL asymp.UCL
##  EA       0.465 0.0403 Inf     0.386     0.544
##  TN       0.683 0.0398 Inf     0.605     0.761
##
## affect = neu:
##  Culture emmean     SE  df asymp.LCL asymp.UCL
##  EA      -0.943 0.0403 Inf    -1.022    -0.864
##  TN      -0.720 0.0398 Inf    -0.798    -0.641
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## $contrasts
## affect = han:
##  contrast estimate     SE  df z.ratio p.value
##  EA - TN     0.442 0.0264 Inf  16.787  <.0001
##
## affect = hap:
##  contrast estimate     SE  df z.ratio p.value
##  EA - TN    -0.218 0.0264 Inf  -8.287  <.0001
##
## affect = neu:
##  contrast estimate     SE  df z.ratio p.value
##  EA - TN    -0.224 0.0264 Inf  -8.485  <.0001
##
## Degrees-of-freedom method: asymptotic
```

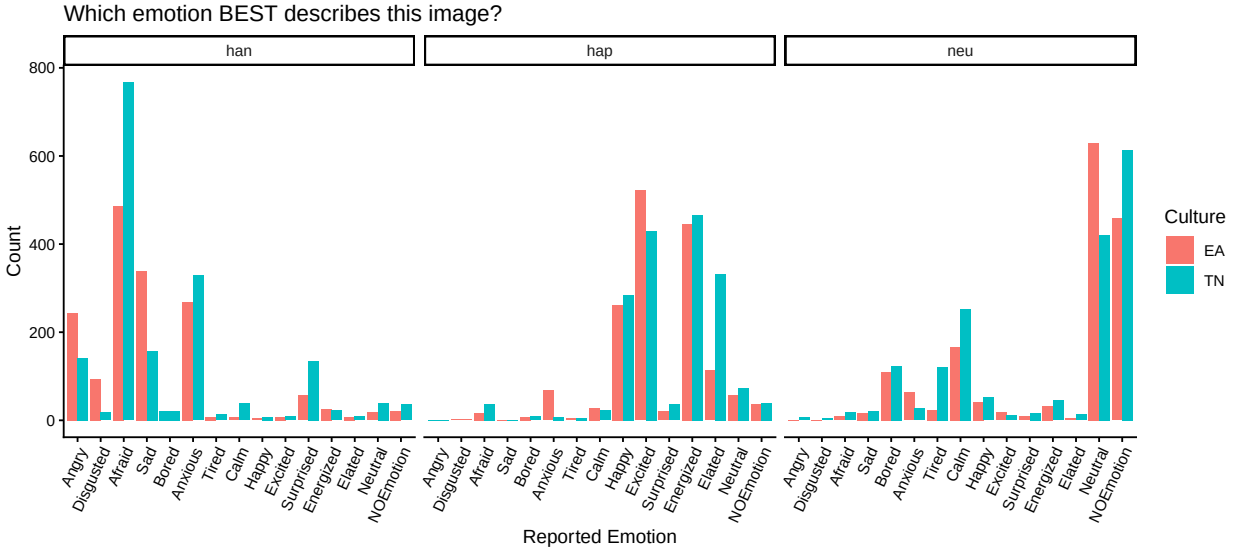
```r
print("ip arousals"); r2(arousal_by_imagetype_ip); r2(arousal_by_imagetype_culture_ip)
```

```
## [1] "ip arousals"

## # R2 for Mixed Models
##
##   Conditional R2: 0.387
##      Marginal R2: 0.367

## # R2 for Mixed Models
##
##   Conditional R2: 0.412
##      Marginal R2: 0.391
```

Discrete Emotions

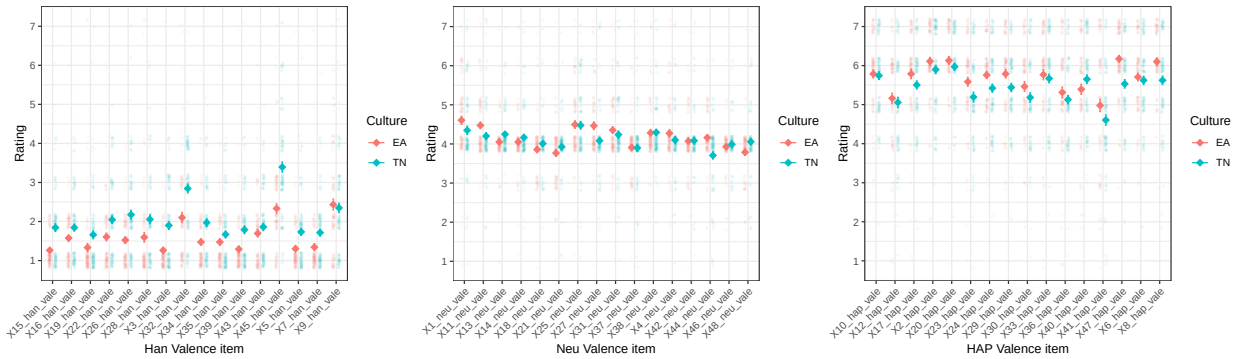
Discrete Emotions: 1 Angry 2 Disgusted 3 Afraid 4 Sad 5 Bored 6 Anxious 7 Tired 8 Calm 9 Happy 10 Excited 11 Surprised 12 Energized 13 Elated 14 Neutral 15 No Emotion





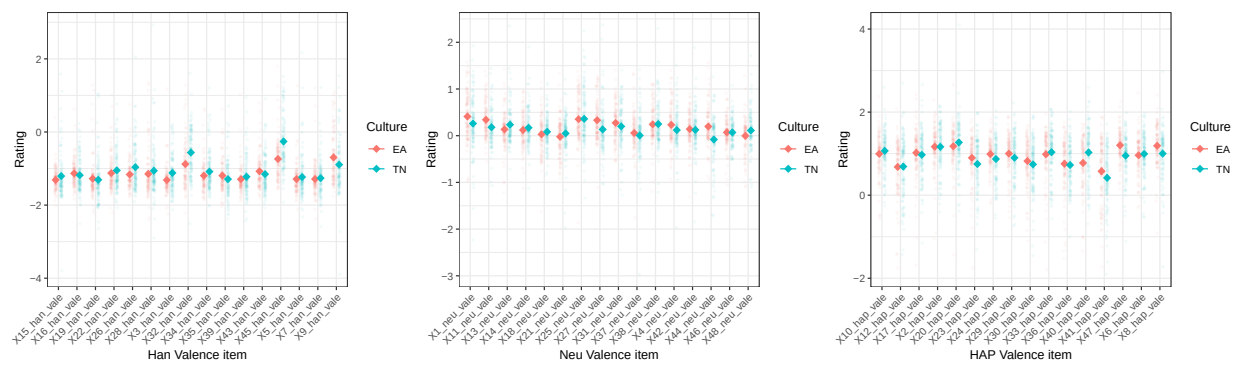
Individual Image Ratings (Raw Valence)

```
p1 + p2 + p3 +
  patchwork::plot_layout(ncol = 3)
```



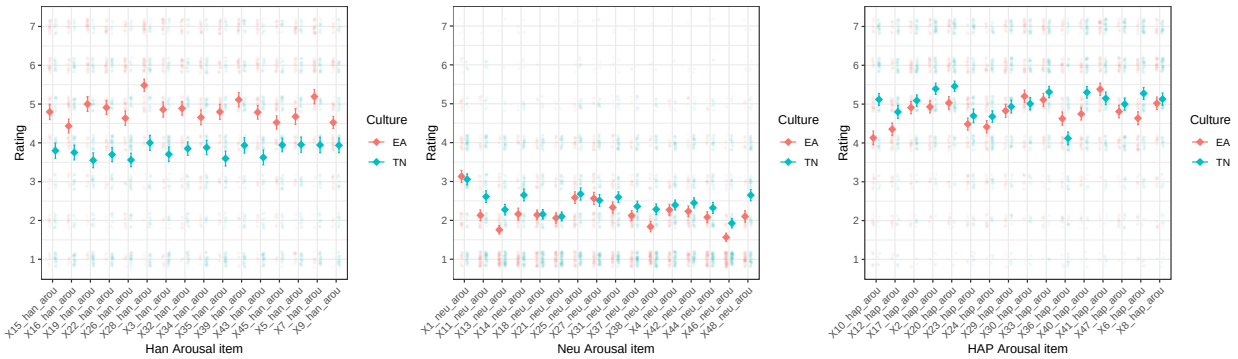
Individual Image Ratings (Ipsatized Valence)

```
p1 + p2 + p3 +
  patchwork::plot_layout(ncol = 3)
```



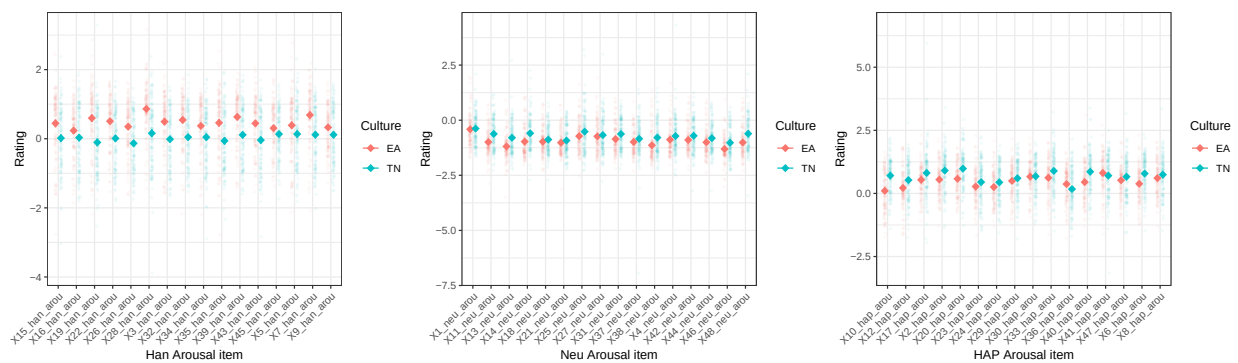
Individual Image Ratings (Raw Arousal)

```
p1 + p2 + p3 +
  patchwork::plot_layout(ncol = 3)
```



Individual Image Ratings (Ipsatized Arousal)

```
p1 + p2 + p3 +
  patchwork::plot_layout(ncol = 3)
```



Create Data Frame: Ratings by image

```
d_image_nonip <- d %>%
  dplyr::select(ResponseId, Culture, X1_neu_vale:X48_neu_desc) %>%
  pivot_longer(cols = X1_neu_vale:X48_neu_desc,
               names_to = "variable",
               values_to = "value") %>%
  dplyr::mutate(image_no = str_extract(variable, "(?<=^X)[0-9]+"),
                image_type = substr(
                  variable, nchar(variable)-7, nchar(variable)-5),
                rating_type = substr(
                  variable, nchar(variable)-3, nchar(variable))) %>%
  dplyr::select(ResponseId, Culture, image_type,
                image_no, rating_type, value) %>%
  pivot_wider(id_cols = c(ResponseId, Culture, image_type, image_no),
              names_from = "rating_type",
              values_from = "value") %>%
  dplyr::mutate(feel = factor(feel,
                              levels = 1:15,
                              labels =
                                c("f_Angry", "f_Disgusted", "f_Afraid",
                                  "f_Sad", "f_Bored", "f_Anxious", "f_Tired",
                                  "f_Calm", "f_Happy", "f_Excited",
                                  "f_Surprised", "f_Energized",
                                  "f_Elated", "f_Neutral", "f_NOEmotion")),
                desc = factor(desc,
                              levels = 1:15,
                              labels =
                                c("d_Angry", "d_Disgusted", "d_Afraid",
                                  "d_Sad", "d_Bored", "d_Anxious", "d_Tired",
                                  "d_Calm", "d_Happy", "d_Excited",
                                  "d_Surprised", "d_Energized",
                                  "d_Elated", "d_Neutral", "d_NOEmotion"))
                ) %>%
  dplyr::mutate(feel = as.character(feel),
                desc = as.character(desc),
                value = 1,
                value2 = 1) %>%
  pivot_wider(id_cols = c(ResponseId, Culture, image_type, image_no,
                          vale, arou, desc, value2),
              names_from = "feel",
              values_from = "value",
              values_fill = 0) %>%
  pivot_wider(id_cols = c(ResponseId, Culture, image_type, image_no,
                          vale, arou,
                          f_Angry, f_Disgusted, f_Afraid, f_Sad,
                          f_Bored, f_Anxious, f_Tired, f_Calm,
                          f_Happy, f_Excited, f_Surprised, f_Energized,
                          f_Elated, f_Neutral, f_NOEmotion),
              names_from = "desc",
              values_from = "value2",
              values_fill = 0) %>%
  dplyr::select(ResponseId:f_NOEmotion,
```

```r
                d_Angry, d_Disgusted, d_Afraid, d_Sad,
                d_Bored, d_Anxious, d_Tired, d_Calm,
                d_Happy, d_Excited, d_Surprised, d_Energized,
                d_Elated, d_Neutral, d_NOEmotion)


d_image_ip <- d_ip %>%
  dplyr::select(ResponseId, Culture, X1_neu_vale:X48_neu_desc) %>%
  pivot_longer(cols = X1_neu_vale:X48_neu_desc,
               names_to = "variable",
               values_to = "value") %>%
  dplyr::mutate(
    image_no = str_extract(variable, "(?<=^X)[0-9]+"),
    image_type = substr(
      variable, nchar(variable)-7, nchar(variable)-5),
    rating_type = paste0(
      substr(variable, nchar(variable)-3, nchar(variable)), "_ip")) %>%
  dplyr::select(
    ResponseId, Culture, image_type, image_no, rating_type, value) %>%
  pivot_wider(id_cols = c(ResponseId, Culture, image_type, image_no),
              names_from = "rating_type",
              values_from = "value") %>%
  dplyr::select(-c("feel_ip", "desc_ip"))

d_image <- merge(d_image_nonip, d_image_ip,
                 by = c("ResponseId", "Culture", "image_type", "image_no"))

rm(d_image_nonip, d_image_ip)
```

```r
d_image_csv <- d_image %>%
  group_by(Culture, image_type, image_no) %>%
  dplyr::mutate(
    n = n(),
    vale = mean(vale, na.rm = T), vale_sd = psych::SD(vale, na.rm = T),
    arou = mean(arou, na.rm = T), arou_sd = psych::SD(arou, na.rm = T),
    vale_ip = mean(vale_ip, na.rm = T), vale_ip_sd = psych::SD(vale_ip, na.rm = T),
    arou_ip = mean(arou_ip, na.rm = T), arou_ip_sd = psych::SD(arou_ip, na.rm = T),
  ) %>%
  dplyr::mutate(
    dplyr::across(contains("f_"),
                  ~ mean(.x) * n())) %>%
  dplyr::mutate(
    dplyr::across(contains("d_"),
                  ~ mean(.x) * n())) %>%
  dplyr::select(Culture, image_type, image_no, n,
                vale, vale_sd, arou, arou_sd,
                vale_ip, vale_ip_sd, arou_ip, arou_ip_sd,
                everything(), -ResponseId) %>%
  dplyr::distinct() %>%
  dplyr::arrange(image_type, image_no, Culture)

#write.csv(d_image_csv, "d_imagetest.csv")
```